Measuring A Nation's Income

Mbak 9515: Economics

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Review

- Microeconomics
  - Study of how households and firms
  - Make decisions, Interact in markets
- Macroeconomics
  - Study of economy-wide phenomena
  - Including inflation, unemployment, and economic growth

Readings

- Chapter 5. Measuring A Nation's Income
  - How to calculate GDP
  - Nominal vs Real

The Economy's Income and Expenditure

- Gross Domestic Product (GDP)
  - Measures the total *income* of everyone in the economy
  - Measures the total *expenditure* on the economy's output of goods and services
- For an economy as a whole
  - Income must *equal* expenditure

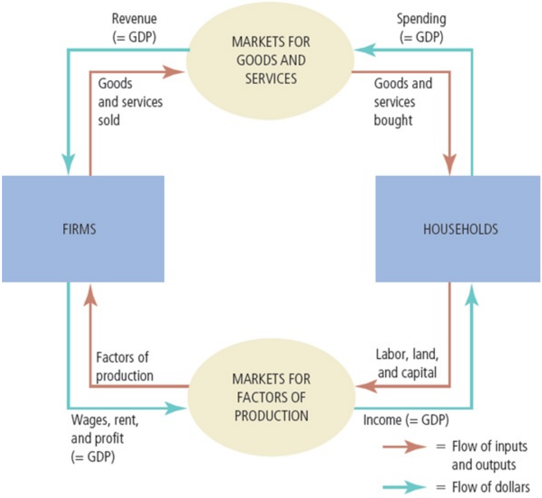
The Economy's Income and Expenditure

Circular-flow diagram

- Assumptions
  - Markets
    - Goods and services
    - Factors of production
  - Households
    - Spend all of their income
    - Buy all goods and services
  - Firms
    - Pay wages, rent, profit to resource owners

The Economy's Income and Expenditure

Circular-flow diagram



Gross Domestic Product

Measuring the Nation's Output

- GDP is *market value of all* *final* goods and services produced *within* a country *in a given period of time*.

Gross Domestic Product

Market Value of all

- Convert production into a dollar value
  - why? Macroeconomists are interested in the behavior of the economy as a whole
- e.g.) Orchardia's GDP
  - 4 apples and 6 bananas
  - 3 pairs of shoes
  - apples sell for \$0.25 each, bananas for \$0.50 each and shoes for \$20.00 a pair
  - 4 apples$\times$\$0.25 + 6 bananas$\times$\$0.50 + 3 shoes$\times$\$20.00 = *\$64*
- It is a convenient way to *add and aggregate* the many different goods and services

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Gross Domestic Product

Final Goods and Services

- Final vs. intermediate goods or services
  - **Final goods or services**: goods or services consumed by the ultimate user
  - they are *counted as* part of GDP
  - **Intermediate goods or services**: goods or services used up in the production of final goods and services
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Gross Domestic Product

Final Goods and Services

- e.g. 1) A farmer grows wheat, which he sells to a miller for \$100. The miller turns the wheat into flour, which he sells to a baker \$150. The baker turns the wheat into bread, which he sells to consumers for \$180. Consumers eat the bread. What is the GDP in this economy?

  - Only final goods (bread in this example) are counted as GDP; **\$180**

Gross Domestic Product

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Gross Domestic Product

Final Goods and Services

- e.g. 2) A farmer produces \$100 worth of milk. He sells \$40 worth of milk to his neighbors and uses the rest to feed his pigs, which he sells to his neighbors for \$120. What is the farmer's contribution to the GDP?
  - milk (\$40) + pig (\$120) = **\$160**; the \$60 worth of milk serves as an intermediate good for the pork

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Gross Domestic Product

Produced within a Country

- **Domestic**: GDP is a measure of economic activity *within* a given country
  - Only production that takes place within the country's borders is counted

- e.g. 1) The market value of *all* cars produced within U.S. borders are included in the GDP of the U.S. *even if they are made in foreign-owned plants*

- e.g. 2) Harley-davidson Motorcycles produced at in Europe are *not counted in* the GDP of the U.S.

Gross Domestic Product

Produced During a Given Period

- Normally calendar year or quarter (three months)

- e.g.) A 20-year-old house is sold to a young family for \$200,000. The family pays the real estate agent a 6% commission, or \$12,000. What is the contribution of this transaction to GDP?

  - Production of the house: 20 years earlier!
  - Only \$12,000 commission fee paid to the real estate agent is counted in current-year GDP.

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Gross Domestic Product

Value Added

- Indirect way to determine the market value of final goods and services
  - Add up the value added by any firm equals the market value of its product or service minus the cost of inputs purchased from other firms

- e.g. 1) A farmer grows wheat, which he sells to a miller for $100. The miller turns the wheat into flour, which he sells to a baker $150. The baker turns the wheat into bread, which he sells to consumers for $180. Consumers eat the bread. What is the GDP in this economy using the value-added method?

- e.g. 2) The miller turns the wheat into flour at the end of 2024, and the baker sells to consumers at January in 2025. What is the GDP in 2024 and 2025?

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Methods for Measuring GDP

Expenditure Method

- Previously, we defined GDP as the quantity of goods and services *produced* by an economy
- Also, we studied *value-add method* to calculate GDP indirectly.

- All current production by firms must be either:
  - Bought by households, other firms, government, and foreigners; or
  - Left unsold is added to inventories and treated as 'spending' or 'bought' by the firm which makes it
- So, GDP can also be measured as the sum of '*expenditure*' on domestic production by households, all firms, government, and foreigners

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Methods for Measuring GDP

Expenditure Method

- Four General Categories
  - Consumption of goods and services (C): purchases by the **households**
  - Private investment of goods and services (I): purchases by **firms/businesses**
  - Government goods and services (G): purchase by **governments**
  - Net Exports (NX) = Exports (X) - Imports (M): net purchases by **foreigners**

- GDP defined as $Y = C + I + G + NX$

- Bureau of Economic Analysis

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Expenditure Method

Consumption

$$Y = C + I + G + NX$$

- Consumption (69%): a final good or service purchased by a household
- **Goods** (24%): a tangible item that consumers gain ownership over when purchasing
  - Durable Goods (9.5%): *long-lived* consumer goods
    - Cars, Furniture, Recreation Goods
  - Non-Durable Goods (14.9%): *shorter-lived* goods
    - Food, Clothing, Gasoline
- **Services** (45%): an intangible good that we purchase but do not gain ownership of
  - health care, airline tickets, financial services, etc.

Expenditure Method

Investment

$$Y = C + I + G + NX$$

- Investment (16.8%): goods and services purchased by firms

  - **Business Fixed Investment** (13.1%): Capital machinery and plants
    - Tractor, Computer, Software, Desk, Building/Plant
  - **Residential Investment** (3.5%): New home construction
  - **Inventory Investment** (or Change in private inventories, 0%): the unsold goods to company inventories

Expenditure Method

Investment

- e.g.) Inventory Investment
  - Best Buy has \$4,000 worth of computers on shelf in 2024
  - Sell \$3,000 in computers in 2024 (Consumption increases by \$3,000)
  - \$1,000 in computers remain unsold (increase in inventory = Investment increases by \$1,000)
  - GDP in 2024 increases by \$4,000 = \$3,000 in C + \$1,000 in I
  - In 2025, remaining \$1,000 in computers are sold (C increases by \$1,000)
  - Change in inventory? Decrease by \$1,000, I = -\$1,000
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Expenditure Method

[Note] House

- Are homes a durable good?
  - **New houses** are not treated as consumer durable but as part of *investment*
  - Households do not take their home when they move
  - Household buying a home $\approx$ Firm purchasing a plant
  - **Rental Housing**: counted as a service
  - Housing Services make up 12% of the total GDP!

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Expenditure Method

Government purchases

$$Y = C + I + G + NX$$

- Government Purchases (17.4%): Goods and services that the government buys
  - **Federal Spending** (6.7%): bought by the Federal Government
    - National Defense (4.0%): fighter planes
    - Non-defense (2.7%): education, training, science
  - **State and Local Spending** (10.6%): bought by local governments
    - highways and roads
  - Does *not* include transfer payments such as Social Security, Unemployment Insurance, Welfare. Why?

Expenditure Method

Net Exports

$$Y = C + I + G + NX$$

- Net Exports (NX) = Exports (X) minus Imports (M) = $X - M$
  - **Exports** (12.9%): domestically produced final goods and services that are sold abroad
  - **Imports** (16%): purchases by domestic buyers of goods and services that were produced abroad
  - Net Exports are 12.9% -16.0% = -3.1% of GDP

$$\begin{aligned} Y &= C + I + G + NX \\ &= 16.4 + 4.3 + 4.0 - 1.0 \\ &= 23.7 \end{aligned}$$

- US GDP in 2nd Quarter 2025 = 23.7 Trillion

GDP using Two Different Methods

Production and Expenditure

- 1,000,000 automobiles valued at \$15,000 each. Of these 700,000 are sold to consumers, 200,000 are sold to businesses, 50,000 are sold to the government, and 25,000 are sold abroad. No automobiles are imported. The automobiles left unsold at the end of the year are held in inventory by the auto producers. Find GDP using the market value of production and expenditure method.

- Production method: 1,000,000×\$15,000=\$15 billion

- Expenditure method:
  - Consumption (C): 700,000×\$15,000=\$10.5 billion
  - Government spending (G): 50,000×\$15,000=\$0.75 billion
  - Net exports (NX): 25,000×\$15,000=\$0.375 billion
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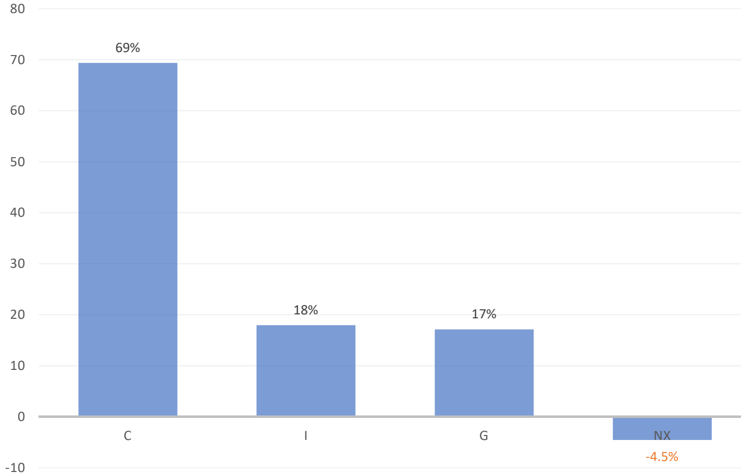
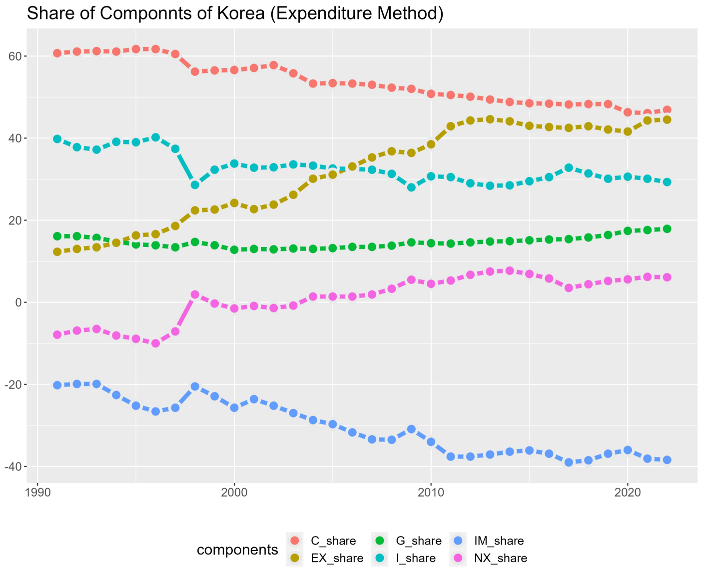
Components of GDP

Percent of Total GDP for the US

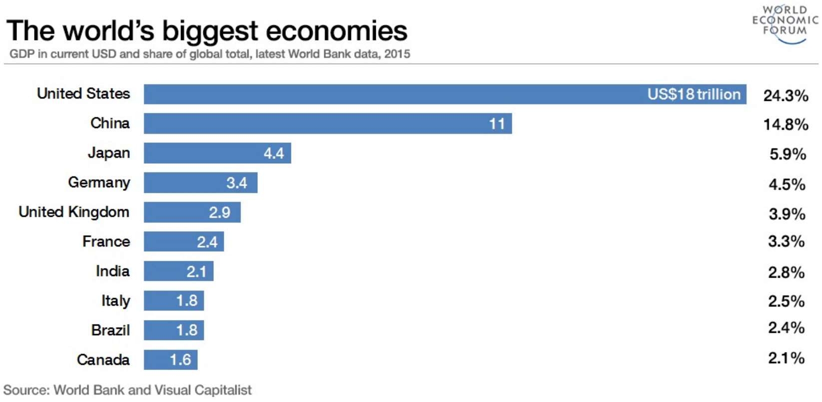


Components of GDP

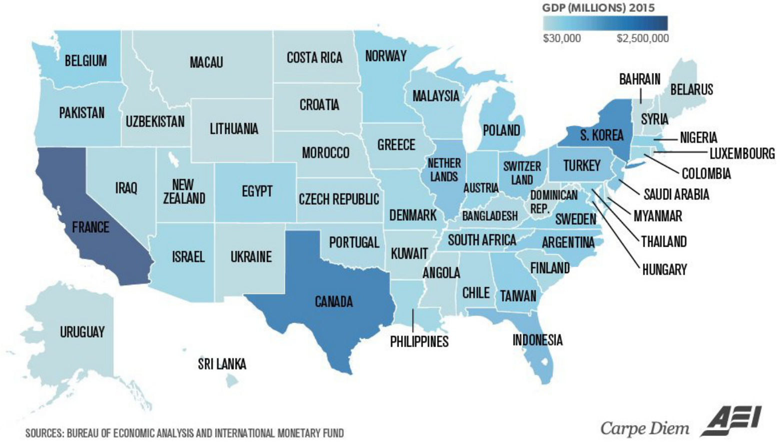
Percent of Total GDP for Korea



GDP by Country



State GDP by Country



Real versus Nominal GDP

- Nominal GDP
  - Production of goods and services
  - Valued at current prices
- Real GDP
  - Production of goods and services
  - Valued at constant prices
  - Designate one year as base year
  - *Not affected* by changes in prices
- For the base year
  - Nominal GDP = Real GDP

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Real versus Nominal GDP

- The GDP deflator
  - Ratio of nominal GDP to real GDP times 100
  - Is 100 for the base year
  - Measures the current level of prices relative to the level of prices in the base year
  - Can be used to take inflation out of nominal GDP ("deflate" nominal GDP)
- Inflation
  - Economy's overall price level is rising
- Inflation rate
  - Percentage change in some measure of the price level from one period to the next

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Nominal vs. Real GDP

	Pizzas		Calzones	
	Quantity	Price	Quantity	Price
2013	10	\$10	15	\$5
2025	20	\$12	30	\$6

- Nominal GDP in 2013: 10×\$10+15×\$5=\$175
- Nominal GDP in 2025: 20×\$12+30×\$6=\$420

- Real GDP in 2013: 10×\$10+15×\$5=\$175
- Real GDP in 2025: 20×\$10+30×\$5=\$350

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2013	10	\$10	15	\$5
2025	20	\$12	30	\$6

- Nominal GDP in 2013: $10 \times \$10 + 15 \times \$5 = \$175$
- Nominal GDP in 2025: $20 \times \$12 + 30 \times \$6 = \$420$

- Real GDP in 2013: $10 \times \$10 + 15 \times \$5 = \$175$
- Real GDP in 2025: $20 \times \$10 + 30 \times \$5 = \$350$

Real GDP and Well-Being

- Real GDP is an imperfect measure of economic well-being
  - Leisure time
  - Nonmarket economic activities
    - parenting and child care services, unpaid housekeeping services, and volunteer services
    - underground economy
    - US GDP in 2018: >$18 trillion but, underground economy≈$2 trillion
  - Environmental quality and resource depletion
    - China's tremendous growth in real GDP but air and water quality
    - *https://www.airvisual.com/earth*
  - Quality of life
    - nice homes, good restaurants and stores; entertainment; high-quality medical services
  - Poverty and economic inequality
- But GDP is related to economic well-being
  - Availability of goods and services
  - Health and education